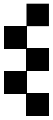
# Reporting Systems



University of Essex

# Reporting Systems

- The purpose of a reporting system is to create meaningful information from disparate data sources and to deliver that information to the proper users on a timely basis.
- Reporting systems differ from data mining because they create information using the simple operations of sorting, filtering, grouping, and making simple calculations.

# RFM Analysis (1)

| RFM Analysis |
| --- |
| • Simple report-based customer classification scheme |
| • Score customers on recentness, frequency, and monetary size of orders |
| • Typically, divide each criterion into 5 groups and score from 1 to 5 |

# RFM Analysis (2)

- RFM analysis is a way of analyzing and ranking customers according to their purchasing patterns.
- It is a simple technique that considers
  - how recently (R score) a customer ordered,
  - how frequently (F score) a customer orders,
  - and how much money (M score) the customer spends per order.

# RFM Analysis (3)

- Customers are sorted into five groups, each containing **20%** of the customers.
- Each group is given a numerical value:
  - **1** = top 20%
  - **2, 3, 4** = each 20% in between top and bottom 20%
  - **5** = bottom 20%
- To produce an RFM score, we need only **two things**: **customer data and sales data** for each purchase (the date of the sale and the total amount of the sale) made by each customer.

# RFM Score Report

Each customer is ranked for **R** (recent), **F** (frequent), and **M** (money) characteristics—1 is highest (best) and 5 is lowest (worst) score

| | RFM Score | | |
|---|---|---|---|
| Customer | R | F | M |
| Able, Ralph | 1 | 1 | 2 |
| Baker, Susan | 2 | 2 | 3 |
| George, Sally | 3 | 3 | 3 |
| Tyler, Jenny | 5 | 1 | 1 |
| Jacobs, Chantel | 5 | 5 | 5 |

# RFM Score Analysis (1)

- The first customer, Ralph Able, has a score of {1 1 2}, which means that he has ordered recently and orders frequently. His M score of 2 indicates, however, that he does not order the most expensive goods.
    - From these scores, the salespeople can surmise that Ralph is a **good customer** but that they **should attempt to up-sell** Ralph to more expensive goods.

# RFM Score Analysis (2)

- Susan Baker (RFM score of {2 2 3}) is above average in terms of how frequently she shops and how recently she has shopped, but her purchases are average in value.
- Sally George (RFM score of {3 3 3}) is truly in the middle.
- Jenny Tyler (RFM score of {5 1 1}) is a problem. Jenny has not ordered in some time, but, in the past, when she did order, she ordered frequently and her orders were of the highest monetary value.
    - These data suggest that Jenny might be going to another vendor. Someone from the sales team should contact her immediately.

# RFM Score Analysis (3)

- However, no one on the sales team should be talking to Chantel Jacobs (RFM score of {5 5 5}).
- She has not ordered for some time, she doesn't order frequently, and when she does order, she only buys inexpensive items and not many of them.